

MH1200 Linear Algebra I.

Problem set #6: SOLUTIONS

Problem 1:

For each of the following matrices, compute the determinant using cofactor expansion along some row or some column.

$$\begin{aligned} (a) \quad & \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}, & (b) \quad & \begin{pmatrix} -1 & 3 & -4 \\ 2 & 4 & 1 \\ -4 & 2 & -9 \end{pmatrix} \\ (c) \quad & \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}, & (d) \quad & \begin{pmatrix} 1 & 0 & 0 & 1 \\ 1 & 2 & 0 & 0 \\ 1 & 2 & 3 & 0 \\ 1 & 2 & 3 & 4 \end{pmatrix}. \end{aligned}$$

Solution

(a) Cofactor expansion along the 1st row gives

$$\begin{vmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix} = (-1)^{1+2}1 \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} + (-1)^{1+3}1 \begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix} = -1(0-1) + 1(1-0) = 1+1=2$$

(b) Cofactor expansion along 1st row gives

$$\begin{aligned} \begin{vmatrix} -1 & 3 & -4 \\ 2 & 4 & 1 \\ -4 & 2 & -9 \end{vmatrix} &= (-1)^{1+1}(-1) \begin{vmatrix} 4 & 1 \\ 2 & -9 \end{vmatrix} + (-1)^{1+2}3 \begin{vmatrix} 2 & 1 \\ -4 & -9 \end{vmatrix} + (-1)^{1+3}(-4) \begin{vmatrix} 2 & 4 \\ -4 & 2 \end{vmatrix} \\ &= -(-36-2) - 3(-18+4) - 4(4+16) = 0 \end{aligned}$$

(c) Cofactor expansion along the 1st column gives

$$\begin{vmatrix} 2 & 0 & 1 \\ 0 & 2 & -1 \\ 0 & -1 & 2 \end{vmatrix} = (-1)^{1+1}2 \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} = 2(4-1) = 6$$

(d) Cofactor expansion along the 1st row gives

$$\begin{vmatrix} 1 & 0 & 0 & 1 \\ 1 & 2 & 0 & 0 \\ 1 & 2 & 3 & 0 \\ 1 & 2 & 3 & 4 \end{vmatrix} = (-1)^{1+1}1 \begin{vmatrix} 2 & 0 & 0 \\ 2 & 3 & 0 \\ 2 & 3 & 4 \end{vmatrix} + (-1)^{1+4}1 \begin{vmatrix} 1 & 2 & 0 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{vmatrix} = 24 - 0 = 24$$

where we used $\begin{vmatrix} 2 & 0 & 0 \\ 2 & 3 & 0 \\ 2 & 3 & 4 \end{vmatrix} = 2 \cdot 3 \cdot 4 = 24$ (determinant of triangular matrix is product of the diagonal entries) and $\begin{vmatrix} 1 & 2 & 0 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{vmatrix} = 0$ since the 2nd and 3rd rows are identical.

Problem 2:

(from the 2021 midterm.)

Let $\mathbf{M} = [m_{ij}]_{5 \times 5}$ denote the 5×5 matrix which is determined by the following formula

$$m_{ij} = \begin{cases} 2^i & \text{if } i < j \\ 0 & \text{otherwise.} \end{cases}$$

Let $\mathbf{N} = [n_{ij}]_{5 \times 5}$ denote the 5×5 matrix which is determined by the following formula

$$n_{ij} = \begin{cases} x^i & \text{if } j = 1 \\ 0 & \text{otherwise.} \end{cases}$$

(i) Write down the matrices $\mathbf{M}$ and $\mathbf{N}$.

(ii) Calculate the determinant of $\mathbf{M} + \mathbf{N}$.

You may use any standard properties of determinants in your answer.

Solution to (i).

The matrix $\mathbf{M}$ is:

$$\mathbf{M} = \begin{bmatrix} 0 & 2 & 2 & 2 & 2 \\ 0 & 0 & 2^2 & 2^2 & 2^2 \\ 0 & 0 & 0 & 2^3 & 2^3 \\ 0 & 0 & 0 & 0 & 2^4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

And the matrix $\mathbf{N}$ is

$$\begin{bmatrix} x^1 & 0 & 0 & 0 & 0 \\ x^2 & 0 & 0 & 0 & 0 \\ x^3 & 0 & 0 & 0 & 0 \\ x^4 & 0 & 0 & 0 & 0 \\ x^5 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Solution to (ii).

We must calculate

$$\begin{vmatrix} x^1 & 2 & 2 & 2 & 2 \\ x^2 & 0 & 2^2 & 2^2 & 2^2 \\ x^3 & 0 & 0 & 2^3 & 2^3 \\ x^4 & 0 & 0 & 0 & 2^4 \\ x^5 & 0 & 0 & 0 & 0 \end{vmatrix}.$$

To calculate the determinant it is convenient to do a cofactor expansion along the last row. The only non-zero term comes from the $(5, 1)$ term. The corresponding term is:

$$x^5(-1)^{1+5} \begin{vmatrix} 2 & 2 & 2 & 2 \\ 0 & 2^2 & 2^2 & 2^2 \\ 0 & 0 & 2^3 & 2^3 \\ 0 & 0 & 0 & 2^4 \end{vmatrix}.$$

The determinants in this expression is upper triangular so we can use the standard property that its determinant is the product of its diagonal entries. Thus we get:

$$x^5 2 2^2 2^3 2^4 = x^5 2^{10} = 1024x^5.$$

□

Problem 3:

For each of the following matrices, find the values of λ for which $\det(\mathbf{A}) = 0$.

$$(a) \mathbf{A} = \begin{pmatrix} \lambda - 2 & 1 \\ -5 & \lambda + 4 \end{pmatrix}, \quad (b) \mathbf{A} = \begin{pmatrix} \lambda - 4 & 0 & 0 \\ 0 & \lambda & 2 \\ 0 & 3 & \lambda - 1 \end{pmatrix}$$

Solution

$$(a) \quad 0 = \begin{vmatrix} \lambda - 2 & 1 \\ -5 & \lambda + 4 \end{vmatrix} = (\lambda - 2)(\lambda + 4) + 5 = \lambda^2 + 2\lambda - 3 = (\lambda + 3)(\lambda - 1) \Rightarrow \lambda = -3 \text{ or } 1.$$

(b)

$$\begin{aligned} 0 &= \begin{vmatrix} \lambda - 4 & 0 & 0 \\ 0 & \lambda & 2 \\ 0 & 3 & \lambda - 1 \end{vmatrix} = (\lambda - 4) \begin{vmatrix} \lambda & 2 \\ 3 & \lambda - 1 \end{vmatrix} = (\lambda - 4)(\lambda(\lambda - 1) - 6) \\ &= (\lambda - 4)(\lambda^2 - \lambda - 6) = (\lambda - 4)(\lambda - 3)(\lambda + 2) \end{aligned}$$

$$\Rightarrow \lambda = 4 \text{ or } 3 \text{ or } -2.$$

□

Problem 4:

Consider the following two matrices depending on 9 parameters $a, b, c, d, e, f, g, h, i$:

$$\mathbf{A} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} g & h & i \\ 7d - 3g & 7e - 3h & 7f - 3i \\ a + 10d & b + 10e & c + 10f \end{bmatrix}$$

Using the 6 fundamental properties of determinants introduced in the lectures, compute $\det(\mathbf{B})$ in terms of $\det(\mathbf{A})$.

Solution to Problem 3

Notice that the second row of $\mathbf{B}$ is a linear combination of two row vectors:

$$\begin{bmatrix} 7d - 3g & 7e - 3h & 7f - 3i \end{bmatrix} = 7 \begin{bmatrix} d & e & f \end{bmatrix} - 3 \begin{bmatrix} g & h & i \end{bmatrix}$$

So we can start by using the property that determinant is a linear function of specific rows to write

$$\begin{vmatrix} g & h & i \\ 7d - 3g & 7e - 3h & 7f - 3i \\ a + 10d & b + 10e & c + 10f \end{vmatrix} = 7 \begin{vmatrix} g & h & i \\ d & e & f \\ a + 10d & b + 10e & c + 10f \end{vmatrix} - 3 \begin{vmatrix} g & h & i \\ g & h & i \\ a + 10d & b + 10e & c + 10f \end{vmatrix}$$

The second matrix has a repeated row so the determinant is zero. Thus $\det(\mathbf{B})$ equals

$$7 \begin{vmatrix} g & h & i \\ d & e & f \\ a + 10d & b + 10e & c + 10f \end{vmatrix}.$$

Next we use the fact that the third row is a linear combination of two rows to write that $\det(\mathbf{B})$ equals

$$7 \begin{vmatrix} g & h & i \\ d & e & f \\ a & b & c \end{vmatrix} + (7) * (10) * \begin{vmatrix} g & h & i \\ d & e & f \\ d & e & f \end{vmatrix}.$$

The second matrix has a repeated row so it is zero. Thus $\det(\mathbf{B})$ equals

$$7 \begin{vmatrix} g & h & i \\ d & e & f \\ a & b & c \end{vmatrix}.$$

To turn this matrix into $\mathbf{A}$ we need to do three row swaps. Each row swap changed the sign of the determinant. Thus we deduce that $\det(\mathbf{B})$ equals

$$7 * (-1)^3 \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = -7\det(\mathbf{A}).$$

□

Problem 5:

Calculate the following determinant which depends on parameters a , b , and c :

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}.$$

State under which conditions on the parameters this matrix is invertible.

Solution

We'll start by doing row operations on the first column.

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} \xrightarrow{R_2 \rightarrow R_2 - aR_1} \begin{vmatrix} 1 & 1 & 1 \\ 0 & b-a & c-a \\ a^2 & b^2 & c^2 \end{vmatrix} \xrightarrow{R_2 \rightarrow R_2 - a^2R_1} \begin{vmatrix} 1 & 1 & 1 \\ 0 & b-a & c-a \\ 0 & b^2-a^2 & c^2-a^2 \end{vmatrix}$$

Thus the determinant equals: $\begin{vmatrix} b-a & c-a \\ b^2-a^2 & c^2-a^2 \end{vmatrix}.$

Next we notice that we can factor some of these terms nicely:

$$\begin{vmatrix} b-a & c-a \\ (b-a)(b+a) & (c-a)(c+a) \end{vmatrix}.$$

Using the linearity of determinant in columns we can now factor out the factors common to every entry in a column:

$$\begin{vmatrix} b-a & c-a \\ (b-a)(b+a) & (c-a)(c+a) \end{vmatrix} = (b-a) \begin{vmatrix} 1 & c-a \\ (b+a) & (c-a)(c+a) \end{vmatrix} = (b-a)(c-a) \begin{vmatrix} 1 & 1 \\ b+a & c+a \end{vmatrix}.$$

Now note that this matrix results from the row operation $R_2 \rightarrow R_2 + aR_1$ applied to a simpler matrix. Thus the determinant equals:

$$(b-a)(c-a) \begin{vmatrix} 1 & 1 \\ b & c \end{vmatrix} = (b-a)(c-a)(c-b).$$

The matrix will be invertible when its determinant $(b-a)(c-a)(c-b)$ is non-zero. This happens exactly when

$$a \neq b, a \neq c, b \neq c.$$

□

Problem 6:

In this problem we'll think about whether the determinant respects the basic operations on matrices. We discussed the important fact that $\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B})$ in lectures. But what about the other two basic operations?

1. Let $\mathbf{A}$ be a square $m \times m$ matrix and let $r \in \mathbb{R}$. Conjecture a relationship between $\det(r\mathbf{A})$ and $\det(\mathbf{A})$.
2. Use induction to formally prove the relationship you conjectured in Part (i).
3. Is the following formula true? Either prove it or give an example to show it cannot be true.

$$\det(\mathbf{A} + \mathbf{B}) = \det(\mathbf{A}) + \det(\mathbf{B}).$$

Solution to Part 1.

To conjecture a relationship we should try a few examples and think about what happens. For example if we consider the general 3×3 matrix we see

$$\begin{aligned} \det \left(r \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \right) &= \det \left(\begin{bmatrix} ra & rb & rc \\ rd & re & rf \\ rg & rh & ri \end{bmatrix} \right) \\ &= (ra)((re)(ri) - (rf)(rh)) - (rb)((rd)(ri) - (rf)(rg)) + (rc)((rd)(rh) - (re)(rg)) \\ &= r^3(a(ei - fh) - b(di - fg) + c(dh - eg)). \end{aligned}$$

This equals:

$$r^3 \det \left(\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \right)$$

In general we'll conjecture: $\det(r\mathbf{A}) = r^m \det(\mathbf{A})$ where $\mathbf{A}$ is an $m \times m$ matrix.

Solution to Part 2.

We'll prove this by induction on m , the order of the matrix.

The base case of the induction, $m = 1$, is obvious because

$$\det(r[x]) = \det([rx]) = rx = r \det([x]).$$

Now we'll do the induction step. We assume that for all $j < m$ then if $\mathbf{A}$ is a $j \times j$ matrix then $\det(r\mathbf{A}) = r^j \det(\mathbf{A})$.

Let the entries of the matrix $\mathbf{A}$ be denoted as usual by a_{ij} .

We'll calculate the determinant of $r\mathbf{A}$ by doing a cofactor expansion across the first row:

$$\det(r\mathbf{A}) = \sum_{k=1}^m (-1)^{1+k} (ra_{1k}) \det(\text{The } (1, k)\text{-minor of } r\mathbf{A}).$$

Because the $(1, k)$ minor of $r\mathbf{A}$ will by r times the $(1, k)$ minor of $\mathbf{A}$ we rewrite this:

$$\det(r\mathbf{A}) = \sum_{k=1}^m (-1)^{1+k} (ra_{1k}) \det(r(\text{The } (1, k)\text{-minor of } \mathbf{A})).$$

Now we can employ the induction assumption to write:

$$= \sum_{k=1}^m (-1)^{1+k} (ra_{1k}) (r^{m-1} \det(\text{The } (1, k)\text{-minor of } \mathbf{A})).$$

Collecting all the r 's out the front:

$$= r^m \sum_{k=1}^m (-1)^{1+k} a_{1k} \det(\text{The } (1, k)\text{-minor of } \mathbf{A}) = r^m \det(\mathbf{A}),$$

as required.

Solution to Part 3.

This property is not true. For example if we set $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 0 & 0 \\ 1 & 2 \end{bmatrix}$ then

$$\det(\mathbf{A} + \mathbf{B}) = \det\left(\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 1 & 2 \end{bmatrix}\right) = \det\left(\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}\right) = 1.$$

But

$$\det(\mathbf{A}) + \det(\mathbf{B}) = \det\left(\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}\right) + \det\left(\begin{bmatrix} 0 & 0 \\ 1 & 2 \end{bmatrix}\right) = 0 + 0 = 0.$$

□

Problem 7:

Let $\mathbf{A}$ be a 4×4 matrix such that $\det(\mathbf{A}) = 9$. Find

- (a) $\det(3\mathbf{A})$, (b) $\det(\mathbf{A}^{-1})$, (c) $\det(3\mathbf{A}^{-1})$, (d) $\det((3\mathbf{A})^{-1})$.

Solution

(a) $\det(3A) = 3^4 \det(A) = 3^4 \cdot 9 = 729$. (b) $\det(A^{-1}) = \det(A)^{-1} = \frac{1}{9}$. (c) $\det(3A^{-1}) = 3^4 \cdot \det(A^{-1}) = 3^4 \cdot \frac{1}{9} = 9$. (d) $\det((3A)^{-1}) = \det(3A)^{-1} = \frac{1}{729}$.

Problem 8:

Let $\mathbf{A}$ and $\mathbf{B}$ be 3×3 matrices such that

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \mathbf{A} = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \mathbf{B}.$$

1. Describe how $\mathbf{A}$ can be obtained from $\mathbf{B}$ by elementary row operations.
2. If $\det(\mathbf{A}) = 4$, find $\det(\mathbf{B})$.

Solution

1. Define

$$\mathbf{E}_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{pmatrix}, \quad \mathbf{E}_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{E}_3 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{E}_4 = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Then the matrix relation above can be expressed as $\mathbf{E}_1\mathbf{E}_2\mathbf{E}_3\mathbf{A} = \mathbf{E}_4\mathbf{B}$. It follows that

$$\mathbf{A} = \mathbf{E}_3^{-1}\mathbf{E}_2^{-1}\mathbf{E}_1^{-1}\mathbf{E}_4\mathbf{B}$$

This shows that $\mathbf{A}$ can be obtained by performing the following row operations on $\mathbf{B}$:

- Step 1: $R_1 \rightarrow R_1 - 2R_3$ (equivalent to multiplication by $\mathbf{E}_4$).
- Step 2: $R_3 \rightarrow R_3 - 3R_2$ (equivalent to multiplication by $\mathbf{E}_1^{-1}$).
- Step 3: $R_2 \rightarrow \frac{1}{2}R_2$ (equivalent to multiplication by $\mathbf{E}_2^{-1}$).
- Step 4: $R_1 \leftrightarrow R_2$ (equivalent to multiplication by $\mathbf{E}_3^{-1} = \mathbf{E}_3$).

2. We can use the fact that $\mathbf{A}$ is obtained from $\mathbf{B}$ by performing the row operations in steps 1–4 in part 1. We know that the determinant is unchanged under the operations in steps 1 and 2, changes by a factor $1/2$ in step 3, and by a $-$ sign in step 4. Consequently,

$$\det(\mathbf{A}) = -\frac{1}{2}\det(\mathbf{B}).$$

Since $\det(\mathbf{A}) = 4$, it follows that $\det(\mathbf{B}) = -2\det(\mathbf{A}) = -8$.

An alternative solution for part 2:

Taking the determinants of the matrices in the original relation above and using the product theorem for determinants gives

$$\begin{aligned} \det(\mathbf{E}_1\mathbf{E}_2\mathbf{E}_3\mathbf{A}) &= \det(\mathbf{E}_4\mathbf{B}) \Rightarrow \det(\mathbf{E}_1)\det(\mathbf{E}_2)\det(\mathbf{E}_3)\det(\mathbf{A}) = \det(\mathbf{E}_4)\det(\mathbf{B}) \\ &\Rightarrow 1 \cdot 2 \cdot (-1) \cdot \det(\mathbf{A}) = 1 \cdot \det(\mathbf{B}) \\ &\Rightarrow \det(\mathbf{B}) = -2\det(\mathbf{A}) \end{aligned}$$

where we have used $\det(\mathbf{E}_1) = 1$, $\det(\mathbf{E}_2) = 2$, $\det(\mathbf{E}_3) = -1$ and $\det(\mathbf{E}_4) = 1$.

Therefore, if $\det(\mathbf{A}) = 4$ then $\det(\mathbf{B}) = -8$.

□

Problem 9:

(from midterm 2021.)

Consider an $n \times n$ matrix $\mathbf{A}$ and let the vectors corresponding to the rows be represented by

$$\vec{r}_1, \dots, \vec{r}_n.$$

Let i and j be numbers such that $1 \leq i < j \leq n$. (These will refer to two of the rows of $\mathbf{A}$.)

$$\mathbf{A} = \begin{bmatrix} \leftarrow & \vec{r}_1 & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \vec{r}_i & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \vec{r}_j & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \vec{r}_n & \rightarrow \end{bmatrix}.$$

Now let $\mathbf{B}$ denote the matrix you get from $\mathbf{A}$ by replacing

- row i by $\frac{1}{\sqrt{2}}(\vec{r}_i - \vec{r}_j)$,
- and row j by $\frac{1}{\sqrt{2}}(\vec{r}_i + \vec{r}_j)$.

In other words:

$$\mathbf{B} = \begin{bmatrix} \leftarrow & \vec{r}_1 & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \frac{1}{\sqrt{2}}(\vec{r}_i - \vec{r}_j) & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \frac{1}{\sqrt{2}}(\vec{r}_i + \vec{r}_j) & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \vec{r}_n & \rightarrow \end{bmatrix}.$$

Express $\det(\mathbf{B})$ in terms of $\det(\mathbf{A})$. Justify your answer using standard properties of determinants.

Solution

First we do a sequence of two applications of the property that determinants are linear functions of particular rows and columns. If we apply this property to row i we deduce the determinant of $\mathbf{B}$ equals:

$$\frac{1}{\sqrt{2}} \begin{vmatrix} \leftarrow & \vec{r}_1 & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \vec{r}_i & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \frac{1}{\sqrt{2}}(\vec{r}_i + \vec{r}_j) & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \vec{r}_n & \rightarrow \end{vmatrix} - \frac{1}{\sqrt{2}} \begin{vmatrix} \leftarrow & \vec{r}_1 & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \vec{r}_j & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \frac{1}{\sqrt{2}}(\vec{r}_i + \vec{r}_j) & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \vec{r}_n & \rightarrow \end{vmatrix}$$

Now we apply the property to row j in both of these determinants. We get a sum of 4 terms.

$$\begin{aligned}
& \left(\frac{1}{\sqrt{2}}\right)^2 \begin{vmatrix} \leftarrow & \vec{r}_1 & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \vec{r}_i & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \vec{r}_i & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \vec{r}_n & \rightarrow \end{vmatrix} + \left(\frac{1}{\sqrt{2}}\right)^2 \begin{vmatrix} \leftarrow & \vec{r}_1 & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \vec{r}_i & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \vec{r}_j & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \vec{r}_n & \rightarrow \end{vmatrix} \\
& - \left(\frac{1}{\sqrt{2}}\right)^2 \begin{vmatrix} \leftarrow & \vec{r}_1 & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \vec{r}_j & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \vec{r}_i & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \vec{r}_n & \rightarrow \end{vmatrix} - \left(\frac{1}{\sqrt{2}}\right)^2 \begin{vmatrix} \leftarrow & \vec{r}_1 & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \vec{r}_j & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \vec{r}_j & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \vec{r}_n & \rightarrow \end{vmatrix}
\end{aligned}$$

Note that the first and fourth of these terms are zero because they have repeated rows. And note that in the third term the rows i and j appear in the opposite order. When we swap them back into position we pick up an extra minus one which cancels the minus sign on that term. So we get:

$$\left(\frac{1}{\sqrt{2}}\right)^2 \begin{vmatrix} \leftarrow & \vec{r}_1 & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \vec{r}_i & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \vec{r}_j & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \vec{r}_n & \rightarrow \end{vmatrix} + \left(\frac{1}{\sqrt{2}}\right)^2 \begin{vmatrix} \leftarrow & \vec{r}_1 & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \vec{r}_i & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \vec{r}_j & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \vec{r}_n & \rightarrow \end{vmatrix} = \begin{vmatrix} \leftarrow & \vec{r}_1 & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \vec{r}_i & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \vec{r}_j & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & \vec{r}_n & \rightarrow \end{vmatrix}.$$

We conclude $\det(\mathbf{B}) = \det(\mathbf{A})$.

□

Problem 10: (From final exam 2018.)

Consider the following matrix, where $a_1, \dots, a_{16}$ are real constants:

$$\begin{bmatrix} 0 & a_1 & 0 & a_2 & 0 \\ a_3 & a_4 & a_5 & a_6 & a_7 \\ 0 & a_8 & 0 & a_9 & 0 \\ a_{10} & a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & a_{15} & 0 & a_{16} & 0 \end{bmatrix}.$$

What is its determinant? Briefly justify your answer.

Solution

The determinant equals zero.

One way to solve this is to do a direct computation of the determinant. This is easy and fine but it may take a long time.

Another way to solve this problem is to pause and think about it for a bit. This is a 5×5 matrix. But there is a set 4 rows and columns we can see (row 2 and 4 and column 2 and 4) with the property that when we eliminate them from the matrix all the left-over entries are all zero. So if, when we calculate this determinant, at every stage of the cofactor recursion we always choose one of these special rows or columns to do a cofactor expansion along, then in the final stage of the recursion every cofactor will be zero. Thus the determinant will be zero.

□

Problem 11:

(From a 2020 midterm.)

Consider the following matrix $\mathbf{A}$ depending on 4 variables a, b, c and d .

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ a & b & c & d \end{bmatrix}.$$

- (a) What is the determinant of $\mathbf{A}$?
- (b) For what values of the variables is $\mathbf{A}$ invertible?
- (c) In the case that $\mathbf{A}$ is invertible, determine the inverse. (Caution: there are several methods to calculate inverses. Choose your method carefully.)

Solution to (a)

The simplest thing is to do a cofactor expansion down the first row. There is one non-zero term corresponding to the final position. We get:

$$\det(\mathbf{A}) = (-1)^{4+1}a \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = -a.$$

Solution to (b)

A matrix is invertible precisely when its determinant is not equal to zero. In this case: $a \neq 0$.

Solution to (c)

Now assume $a \neq 0$ so the matrix $\mathbf{A}$ is invertible. We'll follow the row operations algorithm to determine the inverse. Note it would be possible and correct to use the adjoint formula, but would be very tedious for this example. Students who chose the adjoint method made more mistakes.

So: following the algorithm we construct the following matrix by putting a 5×5 identity matrix next to $\mathbf{A}$.

$$\left[\begin{array}{cccc|cccc} 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ a & b & c & d & 0 & 0 & 0 & 1 \end{array} \right]$$

Now we do Gauss-Jordan to this matrix to turn the left-half into an identity matrix.

$$\begin{aligned} &\rightarrow \left[\begin{array}{cccc|cccc} a & b & c & d & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \end{array} \right] \rightarrow \left[\begin{array}{cccc|cccc} a & b & c & 0 & 0 & 0 & -d & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \end{array} \right] \\ &\rightarrow \left[\begin{array}{cccc|cccc} a & b & 0 & 0 & 0 & -c & -d & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \end{array} \right] \rightarrow \left[\begin{array}{cccc|cccc} a & 0 & 0 & 0 & -b & -c & -d & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \end{array} \right] \\ &\rightarrow \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & -\frac{b}{a} & -\frac{c}{a} & -\frac{d}{a} & \frac{1}{a} \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \end{array} \right] \end{aligned}$$

The algorithm tells us that the inverse is the square matrix to the right of the partition. Thus the inverse is:

$$\mathbf{A}^{-1} = \left[\begin{array}{cccc} -\frac{b}{a} & -\frac{c}{a} & -\frac{d}{a} & \frac{1}{a} \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

□

Problem 12:**(From final exam 2022.)**

Let $\mathbf{A}$ and $\mathbf{B}$ be 4×4 matrices. Assume that $\mathbf{A}$ is invertible and that $\mathbf{B}$ is obtained from $\mathbf{A}$ by the following sequence of four elementary row operations

- (1) $R_2 \rightarrow R_2 - 2R_3$,
- (2) $R_1 \leftrightarrow R_3$,
- (3) $R_4 \rightarrow 10R_4$,
- (4) $R_3 \rightarrow R_3 + \frac{1}{2}R_4$.

- (a) Can we conclude $\mathbf{B}$ is also an invertible matrix? Justify.
- (b) If so, express $\mathbf{AB}^{-1}$ as a product of elementary matrices.
- (c) Calculate the number

$$\frac{\det(\mathbf{A})}{\det(\mathbf{B})}.$$

Solution to (a).

Yes, $\mathbf{B}$ will also be invertible.

There are many ways to see this using the theorem about different characterizations of invertibility. For example, row operations do not affect the RREF of a matrix, so if $\mathbf{A}$ is invertible, which means its RREF is the identity, then it follows that the RREF of $\mathbf{B}$ is also the identity, so $\mathbf{B}$ is invertible.

An alternative, more concrete way to see this is to note it follows from the theory of elementary matrices that $\mathbf{A}$ and $\mathbf{B}$ are related as follows

$$\mathbf{B} = \mathbf{E}_4 \mathbf{E}_3 \mathbf{E}_2 \mathbf{E}_1 \mathbf{A}$$

where the $\mathbf{E}_i$ are the elementary matrices corresponding to the row operations above, Namely:

$$\mathbf{E}_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{E}_2 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

and

$$\mathbf{E}_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 10 \end{bmatrix}, \quad \mathbf{E}_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1/2 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

We now see that $\mathbf{B}$ is a product of invertible matrices, so it is also invertible.

Solution to (b).

To express $\mathbf{AB}^{-1}$ as a product of elementary matrices we just rearrange the above equation to get:

$$\mathbf{AB}^{-1} = \mathbf{E}_1^{-1}\mathbf{E}_2^{-1}\mathbf{E}_3^{-1}\mathbf{E}_4^{-1}.$$

The inverses of the elementary matrices are obtained in a standard way:

$$\mathbf{E}_1^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{E}_2^{-1} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

and

$$\mathbf{E}_3^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1/10 \end{bmatrix}, \quad \mathbf{E}_4^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1/2 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Solution to (c).

We calculate the number: $\frac{\det(\mathbf{A})}{\det(\mathbf{B})}$.

There are a number of ways of doing this. One way is to use the previous equation.

$$\frac{\det(\mathbf{A})}{\det(\mathbf{B})} = \det(\mathbf{AB}^{-1}) = \det(\mathbf{E}_1^{-1}\mathbf{E}_2^{-1}\mathbf{E}_3^{-1}\mathbf{E}_4^{-1}).$$

Continuing:

$$= \det(\mathbf{E}_1)^{-1}\det(\mathbf{E}_2)^{-1}\det(\mathbf{E}_3)^{-1}\det(\mathbf{E}_4)^{-1} = (1)(-1)(10)^{-1}(1) = -\frac{1}{10}.$$

Another approach is to base your deduction on how the determinant is known to change under elementary row operations.

□

Problem 13:

(From mid-term exam 2022.)

Consider a 6×6 matrix $\mathbf{A} = [a_{ij}]_{6 \times 6}$ whose entries are determined by the formula

$$a_{ij} = \begin{cases} i & \text{if } i + j \leq 7, \\ 0 & \text{otherwise.} \end{cases}$$

- (i) Write down the matrix $\mathbf{A}$.
- (ii) Calculate the determinant of this matrix. Show all steps required to calculate this without using your calculator.
- (iii) What is the reduced row echelon form of this matrix? Briefly justify.

You may use standard theory developed during lectures in your deductions.

Solution to (i)

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 & 2 & 0 \\ 3 & 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 & 0 \\ 5 & 5 & 0 & 0 & 0 & 0 \\ 6 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Solution to (ii)

There are many ways to do this. One way is to repeatedly to cofactor expansion on the far-right column.

Another quick way is to do column swaps to put it into upper-triangular form. Every time you do a column swap you pick up a (-1) .

After 3 column swaps the matrix becomes

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 2 & 2 & 2 & 2 & 2 \\ 0 & 0 & 3 & 3 & 3 & 3 \\ 0 & 0 & 0 & 4 & 4 & 4 \\ 0 & 0 & 0 & 0 & 5 & 5 \\ 0 & 0 & 0 & 0 & 0 & 6 \end{bmatrix}$$

Thus the determinant is:

$$(-1)^3 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = (-1)6!.$$

Solution to (iii)

Because the determinant is non-zero we know that the matrix is invertible. And then we know from the characterizations of invertibility that the RREF is the identity matrix.

□

Problem 14:

Let $\mathbf{A}$ be a 3×3 matrix whose entries are all either 0 or 1. What is the largest value and smallest value of $\det(\mathbf{A})$ (among all 3×3 matrices with entries being 0 or 1 only)?

Solution

The determinant of a general 3×3 matrix $\mathbf{A} = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$ can be calculated to be:

$$\det(\mathbf{A}) = aei + bfg + cdh - ceg - afh - bdi \quad (*)$$

Each term here is either 0 or 1 when the entries of $\mathbf{A}$ are all either 0 or 1. Consider first the problem of finding the largest possible value of $\det(\mathbf{A})$:

If we could have $aei = bfg = cdh = 1$ and $ceg = afh = bdi = 0$ then $(*) = 1+1+1-0-0-0 = 3$ would be the largest possible value. But this cannot happen. To see why not, note that $aei = bfg = cdh = 1$ implies that all the entries of $\mathbf{A}$ are non-zero and therefore $=1$. This in turn implies $ceg = afh = bdi = 1$ and therefore $(*) = 0$. The next best possibility is $aei = bfg = 1$ and $cdh = ceg = afh = bdi = 0$, which gives $(*) = 1 + 1 + 0 - 0 - 0 - 0 = 2$. This possibility can be realized: take $a = e = i = b = f = g = 1$ and $c = d = h = 0$. Thus the largest possible value of $\det(\mathbf{A})$ is 2.

Now consider the problem of finding the smallest possible value of $(*)$. An argument similar to the preceding shows that the smallest value is obtained when $aei = bfg = cdh = ceg = 0$ and $afh = bdi = 1$, in which case $(*) = 0 + 0 + 0 - 0 - 1 - 1 = -2$. This possibility can be realized by taking $a = f = h = b = d = i = 1$ and $c = e = g = 0$. Thus the smallest possible value of $\det(\mathbf{A})$ is -2 .

□

Problem 15:

Consider the following 4×4 *Vandermonde matrix* depending on variables a, b, c, d :

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ a & b & c & d \\ a^2 & b^2 & c^2 & d^2 \\ a^3 & b^3 & c^3 & d^3 \end{pmatrix}$$

Show that $\det(\mathbf{A}) = (b-a)(c-a)(d-a)(c-b)(d-b)(d-c)$. So under which conditions on a, b, c, d is $\mathbf{A}$ invertible?

Solution

This will generalise our answer to Problem 3. We start by doing row operations on column 1. We do: $R_2 \rightarrow R_2 - aR_1$, $R_3 \rightarrow R_3 - a^2R_1$, and $R_4 \rightarrow R_4 - a^3R_1$. We deduce $\det(\mathbf{A})$ equals

$$\begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & b-a & c-a & d-a \\ 0 & b^2-a^2 & c^2-a^2 & d^2-a^2 \\ 0 & b^3-a^3 & c^3-a^3 & d^3-a^3 \end{vmatrix} = \begin{vmatrix} b-a & c-a & d-a \\ b^2-a^2 & c^2-a^2 & d^2-a^2 \\ b^3-a^3 & c^3-a^3 & d^3-a^3 \end{vmatrix}.$$

Note the following identities:

- $b^2 - a^2 = (b - a)(b + a)$
- $b^3 - a^3 = (b - a)(b^2 + ab + a^2)$

Using these we can rewrite this determinant:

$$= \begin{vmatrix} b-a & c-a & d-a \\ (b-a)(b+a) & (c-a)(c+a) & (d-a)(d+a) \\ (b-a)(b^2+ab+a^2) & (c-a)(c^2+ac+a^2) & (d-a)(d^2+ad+a^2) \end{vmatrix}$$

This equals:

$$= (b-a)(c-a)(d-a) \begin{vmatrix} 1 & 1 & 1 \\ b+a & c+a & d+a \\ b^2+ab+a^2 & c^2+ac+a^2 & d^2+ad+a^2 \end{vmatrix}.$$

Now perform the following row operation to simplify this determinant: $R_2 \rightarrow R_2 - aR_1$. This equals:

$$= (b-a)(c-a)(d-a) \begin{vmatrix} 1 & 1 & 1 \\ b & c & d \\ b^2+ab+a^2 & c^2+ac+a^2 & d^2+ad+a^2 \end{vmatrix}.$$

Then $R_3 \rightarrow R_3 - aR_2$ and $R_3 \rightarrow R_3 - a^2R_1$. We get:

$$= (b-a)(c-a)(d-a) \begin{vmatrix} 1 & 1 & 1 \\ b & c & d \\ b^2 & c^2 & d^2 \end{vmatrix}.$$

We calculate this already in Problem 3. Thus we conclude:

$$\det(\mathbf{A}) = (b-a)(c-a)(d-a)(c-b)(d-b)(d-c).$$

The matrix $\mathbf{A}$ is invertible exactly when this determinant is non-zero. This is equivalent to no two of the parameters a, b, c, d being equal.

□

Problem 16:

Generalize the identity in the previous problem to $n \times n$ Vandermonde matrices. Hint: try and write down an induction proof of the identity.

Solution

This is a straightforward generalisation of the previous problem but is technical to write down. The determinant we want to calculate is:

$$\begin{vmatrix} 1 & 1 & 1 & \dots & 1 \\ a_1 & a_2 & a_3 & \dots & a_n \\ a_1^2 & a_2^2 & a_3^2 & \dots & a_n^2 \\ \vdots & \vdots & \vdots & \dots & \vdots \\ a_1^{n-1} & a_2^{n-1} & a_3^{n-1} & \dots & a_n^{n-1} \end{vmatrix}.$$

We will use induction on the order of the matrix to show that this determinant equals

$$\prod_{1 \leq i < j \leq n} (a_j - a_i).$$

We start with elementary row operations on the first column:

$$R_2 \rightarrow R_2 - a_1 R_1, R_3 \rightarrow R_3 - a_1^2 R_1, \dots, R_n \rightarrow R_n - a_1^{n-1} R_1.$$

The result is:

$$\begin{vmatrix} 1 & 1 & 1 & \dots & 1 \\ 0 & a_2 - a_1 & a_3 - a_1 & \dots & a_n - a_1 \\ 0 & a_2^2 - a_1^2 & a_3^2 - a_1^2 & \dots & a_n^2 - a_1^2 \\ \vdots & \vdots & \vdots & \dots & \vdots \\ 0 & a_2^{n-1} - a_1^{n-1} & a_3^{n-1} - a_1^{n-1} & \dots & a_n^{n-1} - a_1^{n-1} \end{vmatrix}.$$

Factoring out common terms we get that the determinant equals $(a_2 - a_1)(a_3 - a_1) \dots (a_n - a_1)$ times

$$\begin{vmatrix} 1 & \dots & \dots & 1 \\ a_2 + a_1 & \dots & \dots & a_n + a_1 \\ \vdots & \dots & \dots & \vdots \\ a_2^{n-2} + a_2^{n-3}a_1 + \dots + a_2a_1^{n-3} + a_1^{n-2} & \dots & \dots & a_n^{n-2} + a_n^{n-3}a_1 + \dots + a_na_1^{n-3} + a_1^{n-2} \end{vmatrix}.$$

We can do row operations (writing down this step is straightforward but quite long to do completely) to simplify this to:

$$(a_2 - a_1)(a_3 - a_1) \dots (a_n - a_1) \begin{vmatrix} 1 & \dots & \dots & 1 \\ a_2 & \dots & \dots & a_n \\ \vdots & \dots & \dots & \vdots \\ a_2^{n-2} & \dots & \dots & a_n^{n-2} \end{vmatrix}.$$

Then using induction we finish with:

$$(a_2 - a_1)(a_3 - a_1) \dots (a_n - a_1) \prod_{2 \leq i < j \leq n} (a_j - a_i) = \prod_{1 \leq i < j \leq n} (a_j - a_i).$$

□

Problem 17:

(From the 2019 midterm.)

- (i) Consider the following matrix depending on parameters $a, \dots, l \in \mathbb{R}$.

$$\mathbf{A} = \begin{bmatrix} a & b & e & f \\ c & d & g & h \\ 0 & 0 & i & j \\ 0 & 0 & k & l \end{bmatrix}.$$

Show that $\det(\mathbf{A}) = \det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) \det\left(\begin{bmatrix} i & j \\ k & l \end{bmatrix}\right)$.

- (ii) More generally, consider the following matrix. Here $\mathbf{X}$ is a square $l \times l$ matrix and $\mathbf{Z}$ is a square $k \times k$ matrix

$$\mathbf{B} = \begin{bmatrix} \mathbf{X} & \mathbf{Y} \\ \mathbf{0}_{k \times l} & \mathbf{Z} \end{bmatrix}.$$

Note that $\mathbf{B}$ has a block of zeroes in the bottom-left corner. A matrix of this form is called “block upper triangular”. Show that for such a matrix:

$$\det(\mathbf{B}) = \det(\mathbf{X}) \det(\mathbf{Z}).$$

Note that for this part of the problem, full marks will only be given for a completely clear and correct proof.

Solution to (i).

The simplest solution to this is just to do a cofactor expansion along the first column. We calculate that $\det(\mathbf{A})$ equals:

$$a \begin{vmatrix} d & g & h \\ 0 & i & j \\ 0 & k & l \end{vmatrix} - c \begin{vmatrix} b & e & f \\ 0 & i & j \\ 0 & k & l \end{vmatrix}$$

This equals:

$$ad \begin{vmatrix} i & j \\ k & l \end{vmatrix} - bc \begin{vmatrix} i & j \\ k & l \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} \begin{vmatrix} i & j \\ k & l \end{vmatrix}$$

as required.

Solution to (ii).

There are different approaches to this. The clearest way to organise the explanation is by induction.

We will prove this by induction on l , where l is the order of $\mathbf{X}$. (Recall that the *order* of a square matrix means the number of columns, which equals the number of rows).

The base case: $l = 1$. In this case the answer is immediate. Let $\mathbf{X} = [x]$. A cofactor expansion down the first column of the given matrix

$$\begin{bmatrix} x & \mathbf{Y} \\ \mathbf{0}_{k \times 1} & \mathbf{Z} \end{bmatrix}$$

immediately equals the required expression $x |\mathbf{Z}| = |\mathbf{X}| |\mathbf{Z}|$.

The induction step. In this case we assume that the identity is true for all cases where the order of the top left matrix $\mathbf{X}$ is strictly less than l .

To write out this proof we'll introduce some notation. Let the entries of $\mathbf{X}$ be denoted in the usual way: $\mathbf{X} = [x_{ij}]_{l \times l}$. And as usual let $\mathbf{M}_{ij}$ denote the (i, j) -minor matrix of $\mathbf{X}$ (that is $\mathbf{X}$ with the i -th row and j -th column deleted). Also let $\mathbf{Y}_i$ denote the matrix $\mathbf{Y}$ after the i -th row is deleted.

Using this notation, we can write down the result of a cofactor expansion of $\det(\mathbf{B})$ along the first column as follows:

$$\sum_{p=1}^l (-1)^{p+1} x_{p1} \begin{vmatrix} \mathbf{M}_{p1} & \mathbf{Y}_p \\ \mathbf{0}_{k \times (l-1)} & \mathbf{Z} \end{vmatrix}.$$

Note that the square matrix $\mathbf{M}_{p1}$ that appears in this expression has order one less than $\mathbf{X}$. (Because it is obtained by cancelling 1 row and 1 column of $\mathbf{X}$.) Thus we may now apply the induction assumption to the determinants appearing in this expression and continue:

$$\begin{aligned} &= \sum_{p=1}^l (-1)^{p+1} x_{p1} |\mathbf{M}_{p1}| |\mathbf{Z}| \\ &= \left(\sum_{p=1}^l (-1)^{p+1} x_{p1} |\mathbf{M}_{p1}| \right) |\mathbf{Z}| \\ &= |\mathbf{X}| |\mathbf{Z}|. \end{aligned}$$

□

Problem 18.

(a) Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ matrices and assume that $\mathbf{A}$ is invertible. Prove that

$$\det(\mathbf{I}_n + \mathbf{AB}) = \det(\mathbf{I}_n + \mathbf{BA}).$$

(b) Let $\mathbf{X}$ be an $n \times n$ matrix satisfying the property that $\mathbf{X}^n = \mathbf{0}_{n \times n}$. Show that

$$\det(\mathbf{I}_n + \mathbf{X}) \neq 0.$$

Solution to (a)

Because $\mathbf{A}$ is invertible, we can freely insert cancelling pairs $\mathbf{AA}^{-1} = \mathbf{I}_n$.

$$\det(\mathbf{I}_n + \mathbf{AB}) = \det(\mathbf{AA}^{-1} + \mathbf{ABAA}^{-1})$$

This equals:

$$\det(\mathbf{A}(\mathbf{I}_n + \mathbf{BA})\mathbf{A}^{-1})$$

Then using multiplicativity of determinant this becomes:

$$\det(\mathbf{A})\det(\mathbf{I}_n + \mathbf{BA})\det(\mathbf{A}^{-1}).$$

We know that: $\det(\mathbf{A})\det(\mathbf{A}^{-1}) = 1$. Thus the determinant equals:

$$\det(\mathbf{I}_n + \mathbf{BA})$$

as required.

Solution to (b)

Observe that:

$$(\mathbf{I}_n + \mathbf{X})(\mathbf{I}_n - \mathbf{X} + \mathbf{X}^2 - \dots + (-1)^{n-1}\mathbf{X}^{n-1})$$

equals

$$\mathbf{I}_n - \mathbf{X} + \mathbf{X}^2 - \dots + (-1)^{n-1}\mathbf{X}^{n-1}$$

plus

$$\mathbf{X} - \mathbf{X}^2 - \dots + (-1)^{n-1}\mathbf{X}^n$$

which equals

$$\mathbf{I} + (-1)^{n-1}\mathbf{X}^n = \mathbf{I}.$$

Thus $\mathbf{I} + \mathbf{X}$ is invertible and so has a non-zero determinant.

□

Problem 19:

Prove that the two definitions of determinant we introduced in lectures:

- via a cofactor expansion
- via a sum over permutations

are equal.

Solution

This was a challenging exercise which would never appear on an exam. Some steps in the deduction below are left for students to understand without much explanation.

We can prove this by induction on n , where n is the order of the matrix. The base case, which is of 1×1 matrices, is clearly true.

Now consider the induction step. We assume the permutation formula equals the usual determinant calculated via cofactor expansion for $(n-1) \times (n-1)$ matrices. We must explain why it works for $n \times n$ matrices as well.

Consider an $n \times n$ -matrix $\mathbf{A}$ with entries a_{ij} . The permutation formula is:

$$\sum_{\substack{\text{permutations} \\ \pi : \{1, \dots, n\} \rightarrow \{1, \dots, n\}}} (\text{sgn}(\pi)) a_{1\pi(1)} \dots a_{n\pi(n)}.$$

Denote the set of permutations of $\{1, \dots, n\}$ by $\text{Perm}(n)$.

Let $1 \leq i \leq n$. Denote by $\text{Perm}(n, i)$ the subset of $\text{Perm}(n)$ consisting of permutations π satisfying the property that $\pi(1) = i$. Note that these subsets decompose $\text{Perm}(n)$ into disjoint subsets:

$$\cup_{i=1}^n \text{Perm}(n, i) = \text{Perm}(n) \quad (\star).$$

Our strategy is to use this decomposition to replace the sum over permutations $\text{Perm}(n)$ with a double sum over $1 \leq i \leq n$ and $\text{Perm}(n, i)$.

Every permutation $\pi \in \text{Perm}(n, i)$ determines a permutation $\pi' \in \text{Perm}(n-1)$ obtained by removing 1 from the domain, $\pi(1)$ from the codomain, and renumbering remaining elements to run from 1 to $n-1$.

The reader can check that if $\pi \in \text{Perm}(n, i)$ then:

$$\text{sgn}(\pi) = (-1)^{i+1} \text{sgn}(\pi').$$

We use the decomposition $(\star)$ to rewrite the permutation formula:

$$\sum_{i=1}^n a_{1i} \left(\sum_{\substack{\text{permutations} \\ \pi : \{1, \dots, n\} \rightarrow \{1, \dots, n\} \\ \text{such that } \pi(1) = i}} (\text{sgn}(\pi)) a_{2\pi(2)} \dots a_{n\pi(n)} \right).$$

By replacing every permutation $\pi : \{1, \dots, n\}$ with its corresponding $\sigma = \pi' : \{1, \dots, n-1\} \rightarrow \{1, \dots, n-1\}$ and replacing $\text{sgn}(\pi) = (-1)^{i+1} \text{sgn}(\pi')$ this can be rewritten:

$$\sum_{i=1}^n (-1)^{i+1} a_{1i} \left(\sum_{\substack{\text{permutations} \\ \sigma : \{1, \dots, n-1\} \rightarrow \{1, \dots, n-1\}}} (\text{sgn}(\sigma)) b_{1\sigma(1)}^i \cdots b_{(n-1)\sigma(n-1)}^i \right)$$

where $\mathbf{B}^i = [b_{kl}^i]_{(n-1) \times (n-1)}$ is the minor matrix obtained from $\mathbf{A}$ by removing row 1 and column i .

By the induction assumption we can rewrite this

$$\sum_{i=1}^n (-1)^{i+1} a_{1i} \det(\mathbf{B}^i).$$

This is just the usual cofactor expansion formula $\det(\mathbf{A})$.

□